

Young Families Engagement Model — Design & Approach

Branch: Retrain_Engagement_Model · **Model:** action_transformer_young_families_v0 **Status:** v0 training (2× Tesla T4, detached) on sri-gpu-12t4 . **Author context:** young-families (“Mothers & babies”) concerts need a different notion of engagement from the standard concert model.

1. Why a new model / profile

The standard concert model was trained on webcam-facing footage and **rewards movement and direct gaze**. For young-families concerts that is wrong:

- Registered adults are **not** looking at the camera; if they are, they are usually *not* engaged in the performance.
- Parents’ attention is often on **mobile performers** or on **their own child**.
- **High movement / arousal** here signals *low* engagement (agitation, chasing a toddler), the inverse of a standard gig.
- Parents are **seated** once the concert starts; a **standing adult** is most likely a **performer** and is excluded from audience scoring.

Design decision: keep the existing model as a “**Standard/Concert**” **profile** and add a “**Young Families**” **profile**, selectable in the GUI. Every engagement model shares the same input contract (543-keypoint MediaPipe sequence → binary head), so switching profiles is just loading a different `state_dict` (+ its label semantics and post-processing rules).

2. Engagement as a *default* — bias & algorithm

The core stance: **presume high engagement and only subtract when an anti-engagement cue is detected**. This is encoded in two complementary layers.

2a. Learned bias (in the model)

- **Class-weighted loss.** `CrossEntropyLoss(weight=[engagement=1.5, disengagement=1.0])` tilts the decision boundary toward *engaged*, so ambiguous poses default to engaged rather than disengaged.
- **Label semantics.** The fine→binary map is re-pointed for this audience — most notably `staring` (long, still fixation) flips from *disengaged* → **engaged**, because sustained attention on the stage is a positive signal for parents.

2b. Rule/feature layer (post-processing, outside the model)

Cues that are **not discrete actions** cannot be learned from the Kinetics action set — they are geometric/temporal features computed from the pose stream and the detector, then used to *subtract* from the default-engaged score:

$$\text{score} = \sum (w_0 + w_m \cdot p_{\text{engaged}}) - \sum_k \lambda_k a_k$$

where p_{engaged} is the model probability, a_k are anti-cue activations (orientation, gaze shift, motion energy, hand-to-face, object-in-hand), w_0 is the **default-engaged prior**, and λ_k their penalties.

3. Action & feature taxonomy

Legend:  covered by the trained action head ·  exists in Kinetics-700 but weak in pose-only / needs face ·  not an action — a pose/gaze/orientation/ arousal/object *feature* handled by the rule layer.

Pro-engagement

Cue	Coverage	Where handled
Clapping hands		action head (<code>applause</code>)
Clicking / snapping fingers		action head (weak); candidate feature
Tapping hands on lap		rhythmic-motion feature
Singing along	 / 	action head + mouth/jaw landmarks (or audio)
Smiling		face channel, not body pose
Mimicking performers (dancing)		action head (<code>dancing</code>)
Still head / long gaze fixation	 → learned flip	<code>staring</code> → engaged + head-pose-variance feature
Seated		posture feature; standing→performer gate

Anti-engagement

Cue	Coverage	Where handled
Moving around a lot / agitation		motion-energy feature (inverted sign)
Touching face	 / 	hand-to-face-proximity feature
Talking		face/audio (unreliable pose-only)
Drinking		action head (<code>eating_drinking</code>)
Eating		action head (<code>eating_drinking</code>)
Distracted by child (holding arm)		body-orientation + arm-pose feature

Turned away from performance	✗	torso/head orientation feature
Sudden synchronous gaze shift (multi-parent)	✗	multi-person temporal feature
Holding objects (phone)	✓	action head (<code>phone_distraction</code>)
Holding objects (bottle/other)	✗	object-in-hand (YOLO detector) feature

Key takeaway: ~half the cues (clapping, dancing, singing, eating, drinking, phone) map to the trained action head; the other half are **features/rules**, not Kinetics actions. Hence the **hybrid** design rather than forcing everything through an action classifier.

4. New features (rule layer)

Computed per registered adult, per rolling window, from the 543-keypoint stream (+ YOLO boxes). All are cheap and run in the live post-processor.

- **Head-pose variance** — low variance over time $\Rightarrow$ sustained fixation $\Rightarrow$ *engaged*.
- **Synchronous gaze shift** — sudden, *shared* head-orientation change across multiple parents $\Rightarrow$ distraction event $\Rightarrow$ *disengaged*.
- **Torso/head orientation** — yaw away from the stage axis $\Rightarrow$ *disengaged*.
- **Motion energy / arousal** — windowed keypoint velocity; high $\Rightarrow$ agitation $\Rightarrow$ *disengaged* (inverted vs standard model).
- **Hand-to-face proximity** — wrist landmark near face box $\Rightarrow$ touching face.
- **Object-in-hand** — YOLO object overlapping the hand (bottle/phone).
- **Seated/standing geometry** — hip/knee vertical arrangement; standing adult $\Rightarrow$ treated as performer, excluded from scoring.

5. Pipeline

```

flowchart TD
    A[Camera / Insta360 feed] --> B[YOLO person detection]
    B --> C[Face ID - registered adults only]
    C --> D{Seated?}
    D -- No, standing --> DP[Tag as performer - exclude]
    D -- Yes --> E[MediaPipe Holistic\n543 keypoints/frame]
    E --> F[Temporal buffer\n64-frame sequence]
    F --> G[Temporal Transformer\nengagement head]
    F --> H[Rule / feature layer]

    subgraph Learned_bias
        G --> G1[p_engaged\nclass-weighted 1.5]
    end

    subgraph Anti_cue_features

```

```

      H → H1[orientation]
    H → H2[gaze variance / sync shift]
      H → H3[motion energy]
      H → H4[hand-to-face]
      H → H5[object-in-hand]
      end

    G1 → I[Default-engaged fusion\nscore = sigma w0 + wm*p - sum lambda*a]
      H1 → I
      H2 → I
      H3 → I
      H4 → I
      H5 → I

    I → J[Per-adult engagement score]
    J → K[Redis publish → Unity / dashboard]

```

6. Model architecture

TemporalTransformer — identical input contract to the standard model, so profiles are hot-swappable.

- **Input:** 64 frames × (543 keypoints × 3) = 64 × 1629
- **d_model:** 256 · **heads:** 8 · **encoder layers:** 4 · **dropout:** 0.1–0.3
- **Head:** mean-pool over time → 256 → 128 → **2** (engagement / disengagement)
- **Params:** ~3.4M (4-layer/256d variant)

```

      flowchart LR
        X[Input\n64 x 1629] → P[Linear proj\n1629 → 256]
        P → PE[+ Positional encoding\n1 x 500 x 256]
        PE → T1[Transformer encoder x4\n8 heads, ff=1024, dropout]
        T1 → MP[Temporal mean-pool\n→ 256]
        MP → C1[Linear 256 → 128]
        C1 → R[ReLU + Dropout]
        R → C2[Linear 128 → 2]
        C2 → S[Softmax\nengagement / disengagement]

```

Layer stack (per encoder block):

```

      flowchart TD
        IN[token seq 64 x 256] → MHA[Multi-Head Self-Attention\n8 heads]
        MHA → AD1[Add and Norm]

```

```

AD1 → FF[Feed-Forward\n256 → 1024 → 256]
      FF → AD2[Add and Norm]
      AD2 → OUT[to next block]

```

7. Training configuration (v0)

Setting	Value
Server	sri-gpu-12t4 (192.168.200.206)
GPUs	physical 0 and 8 (CUDA_VISIBLE_DEVICES=0,8), 2× T4 DDP
Features	data/processed/features_kinetics700 (reused, no re-extraction)
Labels	hierarchical_labels_young_families.json (staring→engagement)
Loss	CrossEntropyLoss(weight=[engagement=1.5, disengagement=1.0])
Batch	32/GPU × 2 = 64 · Epochs 50 · LR 1e-4 · Seq 64
Output	models/action_transformer_young_families_v0/best_model.pth
Run mode	detached (nohup), polled ~every 5 min

The standard model (action_transformer_12gpus_binary_v2_cleaned) is **untouched**; this is a separate, additive model. Future context profiles (Theatre, Schools...) follow the same pattern: a new hierarchical_labels_<profile>.json + launch script + --output-dir , selected in the GUI.

8. Roadmap to v1 (needs data)

- Label real young-families footage (auto-annotate with this pipeline, then manual correction on extracted frames).
- Add the **face channel** (smiling, talking) and optional **audio** (singing).
- Implement and tune the rule-layer penalties λ_k on held-out sessions.
- Validate against current-model baselines before release.